

MID TERM EXAMINATION
FOURTH SEMESTER (BCA) APRIL, 2026
Subject: Introduction to Data Science

Q1. Write short notes on:

- (i) What are different python libraries?
- (ii) Explain the concept of slicing array with suitable example.
- (iii) How linear regression helps in data science?
- (iv) How can we handle missing data?
- (v) Why we need Data Frames? Give an example.

Q2. Explain following:

- a. What are data collection strategies?
- b. What are the factors involve in increasing the evolution of Data Science?
- c. What are different data scientist roles?

Q3. Explain following:

- a. What is pivoting? Write a python code to implement pivot() and pivot table() on data frame.
- b. What is difference between skewness and kurtosis?
- c. What are aggregation functions in pandas

Q4. Explain following:

- a. Here are the runs scored by a Cricket team in a league of 12 matches:
110, 120, 100, 150, 110, 140, 130, 180, 120, 250, 140, 110.

Construct a box plot for the same.

- b. What are different data types available in NumPy? How negative indexing implemented in NumPy.

Q5. Explain following:

- a. Explain two types of statistics with an example.
- b. How central tendencies and dispersion provides benefits to Data Science?
- c. Explain the data science lifecycle with a diagram.

END TERM EXAMINATION

FOURTH SEMESTER (BCA) MAY–JUNE, 2025

Subject: Introduction to Data Science

Q1. Write short notes of the following:

- a) What is Data Science?
- b) Differentiate between Data Analytics and Data Science?
- c) How is Data Science different from traditional application programming?
- d) Why is Python favored over other languages for Data Science?
- e) How can we handle missing data?
- f) How are Data Science and Machine Learning related to each other?
- g) Why do we need Data Frames? Give an example.
- h) How do we perform indexing and slicing? Give an example.
- i) Does Data Science have power to predict the future? Justify.

Q2 (a) What are the factors involved in the evolution of Data Science?

(b) Discuss the applications of Data Science in various fields.

Q3 (a) What are Data collection strategies?

(b) Elaborate all stages of Data Pre-processing in Data Science.

Q4 (a) Define the following (Any Three):

- i) Skewness and Kurtosis
- ii) Box plots
- iii) Pivot table
- iv) Linear regression

(b) How do central tendencies and dispersion provide benefits to Data Science?

Q5 (a) Explain two types of statistics with an example.

(b) How do matplotlib and seaborn describe the functionality of statistics in Data Science?

Q6 (a) Explain the following libraries (Any Three):

- i) Pandas
- ii) Numpy
- iii) Scipy
- iv) TensorFlow

(b) Write a code to upload a CSV file using pandas and Data Frames.

Q7. a) How forecasting demand can be analyzed in Data Science?

b) What are different patterns in data? Elaborate with an example?

Q8.

a) Differentiate between categorical features and numerical feature with help of example?

b) Perform case study on investigating clinical data by the following tasks (Assume values):

- i) Load the data
- ii) Feature selection the dataset
- iii) Censoring

END TERM EXAMINATION
FOURTH SEMESTER (BCA) JUNE 2024
Subject: Introduction to Data Science

Q1. Answer the following:

- a) What are different Python Libraries?
- b) Explain the concept of slicing array with suitable example.
- c) What are different data security issues in Data Science?
- d) How linear regression helps in Data Science?
- e) How data frames are created from dictionary of series?

UNIT – I

Q2

- (a) What are different data pre-processing steps?
- (b) Explain different applications of Data Science.

Q3

- (a) Explain different data collection strategies.
- (b) What are different Data Science roles?

UNIT – II

Q4 What is the role of descriptive statistics in Data Science? Explain with suitable examples.

Q5 (a) Here are the runs scored by a Cricket team in a league of 12 matches:

110, 120, 100, 150, 110, 140, 130, 180, 120, 250, 140, 110.

Construct box plot for the same.

- (b) What is the difference between skewness and kurtosis?

UNIT – III

Q6 (a) Write a code to implement different statistical functions in Pandas.

(b) What is pivoting? Write a Python code to implement `pivot()` and `pivot_table()` on DataFrame.

Q7 (a) Explain the concept of broadcasting array in NumPy with suitable examples.

(b) What are different data types available in NumPy? How negative indexing is implemented in NumPy?

UNIT-IV

Q8. What is a role of data science in investigating clinical data? Explain with suitable examples.

Q9. What is use of data science in forecasting demands? Explain with suitable examples.

END TERM EXAMINATION
FOURTH SEMESTER (BCA) JULY 2023
Subject: Introduction to Data Science

Q1. Attempt any five parts:

- a) What is a Series and how is it different from a 1-D array, a list, and a dictionary.
- b) Differentiate Supervised and Unsupervised learning techniques.
- c) Briefly explain Bar and whisker plot with example.
- d) Give use of `pivot_table()` in pandas by giving example.
- e) Specify any two python Libraries and their applications.
- f) Explain skewness and its importance.
- g) Given a DataFrame with rainfall data (day of the week and rainfall inches), write code to find the minimum, maximum, Q1, Q2, Q3, mean and standard deviation of rainfall.

UNIT-I

- Q2.** a) Explain the roles and responsibilities of any six Data Science professionals.
b) Explain four major tasks in data pre-processing.

- Q3.** a) What is Data Science Lifecycle? Explain all stages with diagram.
b) Describe any five data collection strategies.

UNIT-II

- Q4.** a) What are the various types of Data in Statistics? Explain with example.
b) Explain and give formula for the following Measures of Variability of data:
i) Standard Deviation
ii) Inter-quartile range
iii) Mean Absolute Deviation
iv) Variance

- Q5.** a) Explain linear regression and write the code to plot true line by taking an example.
b) List various types of graph/chart available in the pyplot of matplotlib library for data visualization. Explain any two of them in brief.

UNIT-III

- Q6.** Give 4 ways of creating NumPy arrays.

Give the code or syntax to perform the following operation on two 2D NumPy arrays array1 and array2 and 1D array array3:

- a) Add array1 and array2.
- b) Find sum of array1 elements over a given axis.
- c) Find product of array2 elements over a given axis.
- d) Change the dimension of array3 to 2D.
- e) Transpose the array created in part d.
- f) Display 2 rows and third column of 2D array array1.

- g) Join two 2D array along row.
- h) Convert array2 to 1D array.
- i) Split array1 into multiple subarrays.

Q7. Give 4 ways of creating Series.

Write Python code to create the following Series:

101 Harsh
 102 Arun
 103 Ankur
 104 Harpal
 105 Divya
 106 Jeet

- a) Show details of 1st 3 employees using head function.
- b) Show details of last 3 employees using tail function.
- c) Show details of 1st 3 employees without using head function.
- d) Show details of last 3 employees without using tail function.
- e) Show value of index no 102.
- f) Show 2nd to 4th records.
- g) Show values of index-no = 101, 103, 105.
- h) Show details of "Arun".

UNIT-IV

Q8.

Give 4 ways to create DataFrame.

Create a DataFrame to store data for 10 students:

Name	Age	Semester I marks out of 600	Semester II marks out of 500	Attendance

Write program to perform following operations on above DataFrame:

- a) Display details of students who scored more than 560 marks in semester I.
- b) Display details of students who scored less than 250 marks in semester II.
- c) Display details of student who scored minimum marks in semester II.
- d) Display details of student who scored maximum marks in semester II.
- e) Display details of students whose attendance is more than 75.
- f) Display details of students whose attendance is less than 50.
- g) Insert 2 new records in DataFrame.
- h) Add a new column corresponding to percentage of marks of both semesters.
- i) Add a new column corresponding to grades:

Both sem percentage	Grade
>=90	O
>=75 to <90	A+
>=60 to <75	A
>=50 to <60	B+
>=40 to <50	B
<40	F

Q9. Create the following DataFrame Sales containing year-wise sales figures for five salespersons in INR. Use the years as column labels, and salesperson names as row labels.

Name	2014	2015	2016	2017
Madhu	100.5	12000	20000	50000
Kusum	150.8	18000	50000	60000
Kinshuk	200.9	22000	70000	70000
Ankit	30000	30000	10000	80000
Shruti	40000	45000	12500	90000

- Display the row labels of Sales.
- Display the column labels of Sales.
- Display the dimensions, shape, size and values of Sales.
- Display the last two rows of Sales.
- Display the first two columns of Sales.
- Change the DataFrame Sales such that it becomes its transpose.
- Add data to Sales for salesman Sumeet where the sales made are [196.2, 37800, 52000, 78438] in the years [2014, 2015, 2016, 2017] respectively.
- Delete the data for the year 2014 from the DataFrame Sales.
- Update the sale made by Shruti in 2017 to 100000.
- Write the values of DataFrame Sales to a comma-separated file SalesFigures.csv on the disk. Do not write the row labels and column labels.
- Change the name of the salesperson Ankit to Vivaan and Kinshuk to Shailesh.
- Delete the data for salesman Madhu from the DataFrame Sales.

IMPORTANT & REPEATED QUESTIONS

UNIT – I (PYQs ONLY)

Most Repeated / Important

1. What are Data collection strategies?
2. Explain different data pre-processing steps.
3. Discuss the applications of Data Science in various fields.

Also Asked (Important)

6. What are different Data Science roles?
7. What are the factors involved in the evolution of Data Science?

UNIT – II (PYQs ONLY)

Most Repeated / Important

1. Explain two types of statistics with an example.
2. Define: Skewness and Kurtosis.
3. Define: Box plots.
4. Define: Linear regression.

UNIT – III (PYQs ONLY)

Most Repeated / Important

1. Write a code to implement different statistical functions in Pandas.
2. What is pivoting? Write a Python code to implement `pivot()` and `pivot_table()` on DataFrame.
3. Explain the following libraries: Pandas, NumPy, SciPy, TensorFlow (Any Three).

Also Asked (Important)

4. Explain the concept of broadcasting array in NumPy with suitable examples.
5. What are different data types available in NumPy? How is negative indexing implemented in NumPy?
6. Write a code to upload a CSV file using pandas and Data Frames.

INTRODUCTION TO DATA SCIENCE

BCA 232 | 4th Semester | IP University (GGSIPU)

PYQ-Based Important Questions — End Term Exam | 60 Marks

LEGEND & COLOUR CODE

- ★ **MOST ASKED** Appeared in 2+ PYQ papers — Extremely high exam probability
- ◆ **IMPORTANT** Appeared at least once or high syllabus weight
- × **4 Papers** Appeared in ALL 4 papers — Don't skip!
- × **3 Papers** Appeared in 3 papers — Very high priority
- × **2 Papers** Appeared in 2 papers — High priority
- × **1 Paper** Appeared once — Prepare if time allows

PYQ PAPERS ANALYSED (4 PAPERS)

END TERM EXAMINATION	FOURTH SEMESTER (BCA)	JULY 2023	Subject: Introduction to Data Science
END TERM EXAMINATION	FOURTH SEMESTER (BCA)	JUNE 2024	Subject: Introduction to Data Science
END TERM EXAMINATION	FOURTH SEMESTER (BCA)	MAY–JUNE 2025	Subject: Introduction to Data Science
MID TERM EXAMINATION	FOURTH SEMESTER (BCA)	APRIL 2026	Subject: Introduction to Data Science

UNIT I — DATA SCIENCE FUNDAMENTALS & PRE-PROCESSING

- Q1.** What are different Data Collection Strategies? Explain each with suitable examples. ★ **MOST ASKED** [10 Marks] ×4 Papers 2023 | 2024 | 2025 | 2026-Mid
- Q2.** Explain all stages of Data Pre-processing in detail. ★ **MOST ASKED** [10 Marks] ×4 Papers 2023 | 2024 | 2025 | 2026-Mid
- Q3.** Discuss the applications of Data Science in various fields. ★ **MOST ASKED** [10 Marks] ×2 Papers 2024 | 2025
- Q4.** What are different Data Science roles? Explain each briefly. ★ **MOST ASKED** [5 Marks] ×3 Papers 2023 | 2024 | 2026-Mid
- Q5.** What is Data Science? How is it different from traditional application programming? ★ **MOST ASKED** [5 Marks] ×2 Papers 2024 | 2025
- Q6.** What is the Data Science Lifecycle? Explain all stages with a suitable diagram. ★ **MOST ASKED** [5 Marks] ×2 Papers 2023 | 2026-Mid
- Q7.** What are the factors involved in the evolution of Data Science? ★ **MOST ASKED** [5 Marks] ×2 Papers 2025 | 2026-Mid
- Q8.** What are different data security issues in Data Science? ◆ **IMPORTANT** [2 Marks] ×1 Paper 2024
- Q9.** How are Data Science and Machine Learning related to each other? ◆ **IMPORTANT** [2 Marks] ×1 Paper 2025
- Q10.** Does Data Science have the power to predict the future? Justify. ◆ **IMPORTANT** [2 Marks] ×1 Paper 2025
- Q11.** Differentiate between Data Analytics and Data Science. ◆ **IMPORTANT** [2 Marks] ×1 Paper 2025

UNIT II — PYTHON, NUMPY, PANDAS & STATISTICS

- Q1.** What is the difference between Skewness and Kurtosis? Explain their types with examples. ★ MOST ASKED
ASKED [5 Marks] ×4 Papers 2023 | 2024 | 2025 | 2026-Mid
- Q2.** What is Linear Regression? Explain Simple and Multiple Linear Regression with the basic equation $y = mx + c$. ★ MOST ASKED [5 Marks] ×4 Papers 2023 | 2024 | 2025 | 2026-Mid
- Q3.** Why do we need DataFrames? How is a DataFrame created from a dictionary of Series? Give an example. ★ MOST ASKED [5 Marks] ×4 Papers 2023 | 2024 | 2025 | 2026-Mid
- Q4.** Explain the concept of array slicing in NumPy with suitable examples (1D and 2D arrays). ★ MOST ASKED
[5 Marks] ×3 Papers 2023 | 2024 | 2025
- Q5.** How do central tendencies and dispersion provide benefits to Data Science? Explain with examples. ★
MOST ASKED [5 Marks] ×3 Papers 2023 | 2025 | 2026-Mid
- Q6.** Explain two types of statistics (Descriptive and Inferential) with suitable examples. ★ MOST ASKED [10 Marks] ×2 Papers 2025 | 2026-Mid
- Q7.** What is a Box Plot? Explain its components. Construct a box plot for: 110, 120, 100, 150, 110, 140, 130, 180, 120, 250, 140, 110. ★ MOST ASKED [5 Marks] ×2 Papers 2024 | 2026-Mid
- Q8.** What is Pivoting in Pandas? Write Python code to implement pivot() and pivot_table() on a DataFrame. ★ MOST ASKED [10 Marks] ×3 Papers 2023 | 2024 | 2026-Mid
- Q9.** Write Python code to implement different statistical functions in Pandas (min, max, mean, median, mode, sum, count, std, var). ★ MOST ASKED [10 Marks] ×2 Papers 2023 | 2024
- Q10.** What are different data types available in NumPy? How is negative indexing implemented in NumPy? ★
MOST ASKED [5 Marks] ×2 Papers 2024 | 2026-Mid
- Q11.** How can we handle missing data in a dataset? ★ MOST ASKED [2 Marks] ×2 Papers 2025 | 2026-Mid
- Q12.** Explain Variance, Standard Deviation, IQR, and Mean Absolute Deviation with formulas. ♦ IMPORTANT
[5 Marks] ×1 Paper 2023
- Q13.** How do Matplotlib and Seaborn describe the functionality of statistics in Data Science? ♦ IMPORTANT
[5 Marks] ×1 Paper 2025
- Q14.** What are different Python libraries used in Data Science? Explain any three briefly. ♦ IMPORTANT [5 Marks] ×2 Papers 2024 | 2026-Mid
- Q15.** What are aggregation and group-by functions in Pandas? Explain with examples. ♦ IMPORTANT [5 Marks] ×1 Paper 2026-Mid
- Q16.** What are the various types of Data in Statistics? Explain with example. ♦ IMPORTANT [2 Marks] ×1 Paper 2023
- Q17.** Why is Python favored over other languages for Data Science? ♦ IMPORTANT [2 Marks] ×1 Paper 2025
- Q18.** What is NumPy? Why is it preferred over Python lists for Data Science? ♦ IMPORTANT [2 Marks] ×1 Paper 2025
- Q19.** What is a Series in Pandas? How is it different from a 1-D array, a list, and a dictionary? ♦ IMPORTANT
[2 Marks] ×1 Paper 2023

UNIT III — NUMPY OPERATIONS, LIBRARIES & INFERENCE STATISTICS

- Q1.** Explain the following Python libraries (Any Three): Pandas, NumPy, SciPy, TensorFlow. ★ MOST ASKED
[10 Marks] ×2 Papers 2025 | 2026-Mid (implied)
- Q2.** Explain the concept of Broadcasting in NumPy with suitable examples. ★ MOST ASKED [5 Marks] ×1 Paper 2024

- Q3.** Write Python code to upload a CSV file using Pandas and perform basic DataFrame operations. ★ **MOST ASKED** [5 Marks] ×1 Paper 2025
- Q4.** Give 4 ways of creating NumPy arrays. Perform operations: add, sum over axis, transpose, reshape, split. ♦ **IMPORTANT** [10 Marks] ×1 Paper 2023
- Q5.** Give 4 ways of creating a Pandas Series. Perform head(), tail(), slicing, and indexing operations. ♦ **IMPORTANT** [10 Marks] ×1 Paper 2023
- Q6.** What is Hypothesis Testing? Explain its steps, Null and Alternative Hypothesis, with an example. ♦ **IMPORTANT** [10 Marks] ×1 Paper Syllabus
- Q7.** What is Correlation Analysis? Explain types of correlation (Positive, Negative, Zero) with examples. ♦ **IMPORTANT** [5 Marks] ×1 Paper Syllabus
- Q8.** Explain the following statistical tests: Z-test, t-test, Chi-Square test, and F-test. ♦ **IMPORTANT** [10 Marks] ×1 Paper Syllabus
- Q9.** What is the Central Limit Theorem (CLT)? State its significance in inferential statistics. ♦ **IMPORTANT** [2 Marks] ×1 Paper Syllabus
- Q10.** Explain Probability Distributions: Poisson, Normal, and Exponential. Give one example each. ♦ **IMPORTANT** [5 Marks] ×1 Paper Syllabus
- Q11.** What are Types of Sampling? Explain any three types with examples. ♦ **IMPORTANT** [2 Marks] ×1 Paper Syllabus
- Q12.** What is Differentiate Supervised and Unsupervised learning techniques? ♦ **IMPORTANT** [2 Marks] ×1 Paper 2023

UNIT IV — CASE STUDIES

- Q1.** What is the role of Data Science in Forecasting Demand? Explain with suitable real-world examples. ★ **MOST ASKED** [10 Marks] ×2 Papers 2024 | 2025
- Q2.** What is the role of Data Science in investigating Clinical Data? Explain with suitable examples. ★ **MOST ASKED** [10 Marks] ×2 Papers 2024 | 2025
- Q3.** What are different patterns in data? Explain each with suitable examples (Trend, Seasonal, Cyclical, Random, Repeating, Cluster). ★ **MOST ASKED** [10 Marks] ×1 Paper 2025
- Q4.** Explain the steps involved in Demand Forecasting using Data Science — from Data Collection to Decision Making. ★ **MOST ASKED** [5 Marks] ×2 Papers 2024 | 2025
- Q5.** Differentiate between categorical features and numerical features with examples. ♦ **IMPORTANT** [5 Marks] ×1 Paper 2025
- Q6.** Perform a case study on investigating clinical data: (i) Load data (ii) Feature Selection (iii) Censoring. ♦ **IMPORTANT** [10 Marks] ×1 Paper 2025

QUICK PRIORITY SUMMARY

TOPIC	UNIT	APPEARED	MARKS
Data Collection Strategies	I	×4	10 M
Data Pre-processing Steps	I	×4	10 M
Skewness & Kurtosis	II	×4	5 M

TOPIC	UNIT	APPEARED	MARKS
Linear Regression	II	×4	5 M
DataFrames (creation & use)	II	×4	5 M
Pivoting (pivot & pivot_table)	II	×3	10 M
Array Slicing	II	×3	5 M
Central Tendency & Dispersion	II	×3	5 M
Data Science Roles	I	×3	5 M
Applications of Data Science	I	×2	10 M
Statistical Functions in Pandas	II	×2	10 M
Box Plot (cricket numerical)	II	×2	5 M
Types of Statistics	II	×2	10 M
Data Science Lifecycle	I	×2	5 M
Demand Forecasting Case Study	IV	×2	10 M
Clinical Data Case Study	IV	×2	10 M
Python Libraries (Pandas/NumPy/SciPy/TF)	III	×2	10 M

All the best Marooof! Focus on ×4 and ×3 topics first — they cover 70% of exam marks.

BCA 232 | Introduction to Data Science | IP University | Batch 2024–2027